

Network Forensics

Module IV

Introduction of Network Forensics

Network forensics is the capture, recording, and analysis of network events in order to discover the source of security attacks or other problem incidents.

Network forensics looks at network traffic, logs, and other data about network use.

It helps solve computer crimes, network problems, and data theft.

The main job of network forensics is to find and keep digital evidence that can be used in court.

By looking at network records, people who solve computer crimes can piece together what happened.

Network forensics investigation methodology

OSCAR framework contains **Obtain**, **Strategize**, **Collect**, **Analyze** and **Report**. Phases in a systematic setting, to define a standard methodology for network forensics.



Obtain

Finding out as much information about the incident and discussing goals for the network forensic exercise, are the primary goals of this phase.

Once an incident takes place, a few key details become equally important such as **Date and Time**, **Systems** and **endpoints affected**, **Presence of Logs**, a narrative from the client's IT team, and the steps taken to contain evidence and ensure continuity of business.

This phase answers the '**What**' and '**Where**' part of the incident.

Strategize

The blueprint/plan of the investigation is to be devised in this phase.

Points to considered while making strategy:

Define clear goals and timelines for the investigation.

Think of the sources of evidence.

Plan timely updates for the client.

Collect

Acquire the evidence as per the plan defined in the previous phase.

Collection of the evidence requires us to document all the systems that were accessed and used; capture and save the data streams to the hard drive, and collect logs from servers, applications, proxies, and firewalls.

The best practices for evidence collection include:

Create copies of the evidence and generate cryptographic hashes for verifiability.

Work on copies of data instead of the original evidence.

Use commercial and industry-standard tools.

Document all your actions and steps taken.

Analyze

Most critical phase includes making sense of the data, deriving a story behind the incident, and determining what happened and caused it.

This is the phase where you start working on evidence by using automated and manual techniques and industry's leading forensic tools to correlate data from numerous sources,

formulate a timeline of events,

eliminate sinkholes, and

derive working theories with supporting evidence.

Report

Reporting phase is also one of the most crucial phases of an investigation as your reports would be read by CEOs, CFOs, CISOs, Jury, Insurance, Legal teams, and many others.

It is vital that the produced report is in a layman's language and easy to follow, and should be well understood by people with no technical expertise.

The report should begin with executive summaries followed by technical findings backed by technical evidence.

Network-based evidence sources

1. Packets transmitted over wired and wireless links
2. Switches
3. Routers
4. DHCP Servers
5. Name Servers
6. Authentication Servers
7. Network IDS/IPS
8. Firewalls
9. Web Proxies
0. Application Servers
11. Central Log Servers

Analysis of network evidences IDS

Analyzing **network evidence from an IDS** is a core task in network forensics.

It helps investigators understand
what happened,
how it happened, and
who was involved during a security incident.

What is Network Evidence in IDS

What is Network Evidence in IDS

Network evidence refers to **data collected or generated by an IDS** when it monitors network traffic.

This evidence is used to detect and investigate malicious activities.

Common IDS sources of evidence

- Alert logs
- Packet captures (PCAP files)
- Flow records (NetFlow/IPFIX)
- Metadata (timestamps, IPs, ports, protocols)

Types of IDS Evidence

a) Alert Logs

- Generated when traffic matches known attack signatures or anomalies
- Includes attack type, severity, source IP, destination IP, and time

b) Packet Data

- Raw network packets captured during suspicious activity
- Useful for deep inspection of payloads

c) Traffic Flow Data

- Summarized communication details (who talked to whom, how long, how much data)

Steps in IDS Network Evidence Analysis

Step 1: Collection of Evidence

- Secure IDS logs and packet captures
- Preserve data integrity (hashing, write protection)

Step 2: Alert Correlation

- Group related alerts
- Remove false positives
- Identify attack patterns (e.g., scanning → exploitation)

Step 3: Traffic Examination

- Analyze packets using tools like Wireshark
- Check payloads for malware, exploits, or commands

Step 4: Timeline Reconstruction

- Arrange events chronologically
- Understand the sequence of the attack

Step 5: Attribution and Impact Analysis

- Identify attacker source (internal/external)
- Determine affected systems and data

Tools Used for IDS Evidence Analysis

Snort / Suricata – IDS alert generation

Wireshark – Packet-level analysis

Zeek (Bro) – Network behavior analysis

Splunk / ELK Stack – Log correlation and visualization

Router investigation

Examining a router to collect, analyze, and preserve evidence related to network activities such as unauthorized access, data theft, malware communication, or cyber-attacks.

Routers control and log network traffic, they are a key source of digital evidence.

Steps in Router Investigation

1. Identify the Router

Determine the make, model, IP address, and location of the router. Identify whether it is a home router, enterprise router, or ISP router. Note connected interfaces (LAN, WAN, wireless).

2. Secure the Router

Ensure the router is isolated from unauthorized access. Avoid rebooting or resetting, as this may erase volatile data. Change admin access carefully if required, while documenting actions.

3. Collect Volatile Data

(Important because it is lost after reboot)

Routing tables

ARP tables

Active connections

NAT tables

Current memory (RAM) data, if supported

4. Collect Non-Volatile Data

Configuration files

Startup configuration

Firmware version

Access Control Lists (ACLs)

Stored logs and user credentials (if legally permitted)

5. Analyze Router Logs

Examine system logs for:

Unauthorized access attempts

Configuration changes

Login timestamps

Traffic anomalies

Look for evidence of attacks like:

DDoS, Spoofing, Malware communication

6. Examine Network Traffic

Analyze traffic passing through the router.

Identify:

Suspicious IP addresses

Unusual ports or protocols

Data exfiltration attempts

Use packet capture tools if available.

7. Check for Configuration Tampering

Verify firewall rules and port forwarding settings.

Look for:

Backdoors

Unauthorized VPNs

Modified DNS settings

8. Correlate Evidence

Match router data with:

Server logs

Firewall logs

IDS/IPS alerts

Build a timeline of events.

9. Document Findings

Record:

All steps taken

Tools used

Evidence collected

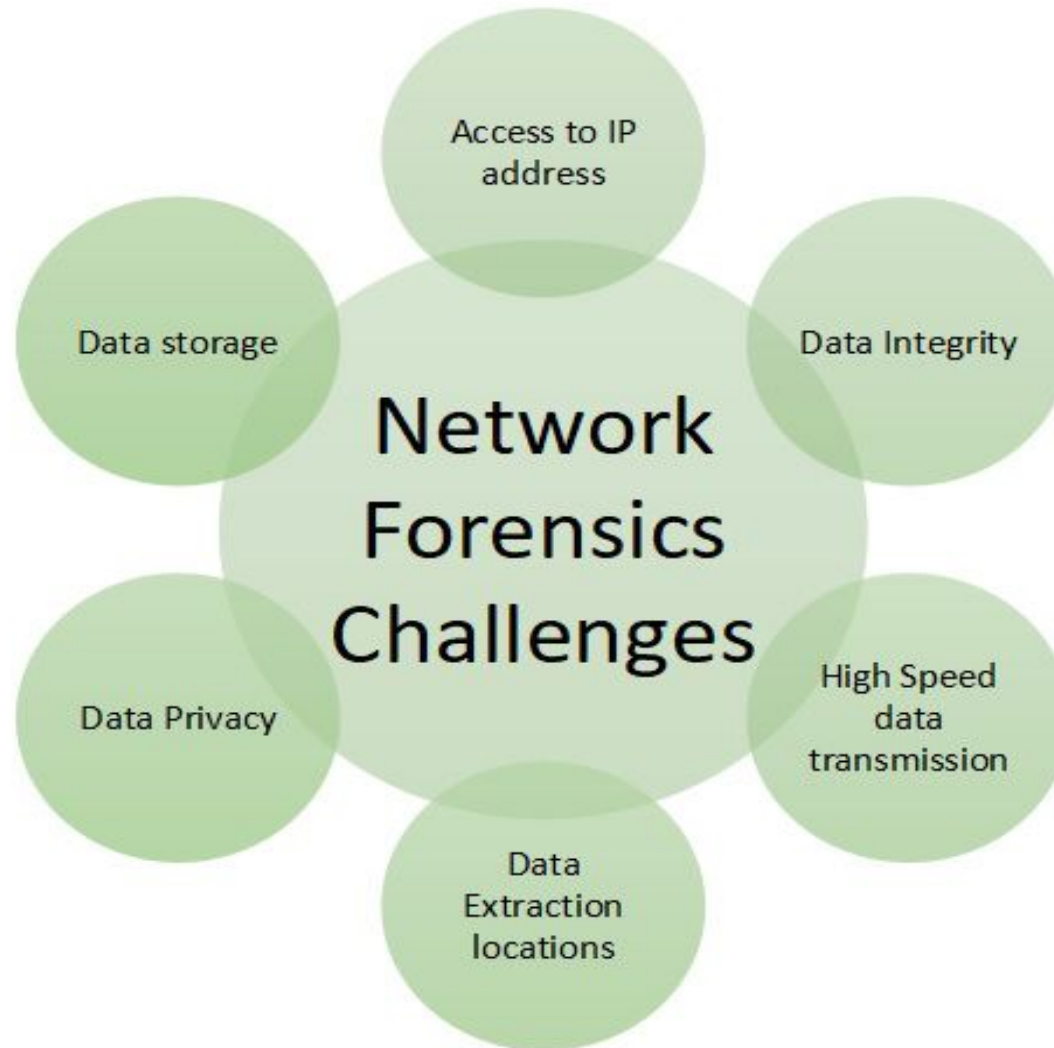
Maintain chain of custody for legal admissibility.

10. Report and Preserve Evidence

Prepare a clear investigation report.

Securely store router data and backups for future reference or court use.

Challenges of Network Forensics



Data Storage: Modern network speeds create massive amounts of data, making full packet capture and analysis costly, resource-intensive, and difficult to manage.

Data Integrity: data integrity should be maintained using the end-to-end approach. It means that the use of software, as well as the hardware, should be seamless.

Access to IP Addresses: Spoofing the IP address is one such technique in which the intruder can show a fake IP address to the devices registered on the network. Spoofing is the technique that is mostly used in DDoS attacks

High-Speed Data Transmission: Millions of data packets are transmitted on networks within a short period, and these packets pass through a vast number of interconnected devices.

Location of Data Extraction: Difficult for forensic experts to identify the device and appropriate location for extracting data.

Data Privacy: Cryptographic tools can be used for this purpose, which may include BBS (Boneh–Boyen–Shacham) short group signatures and group signatures.

Means that anyone of the group anyone of the group members can create a signature, which is verifiable by the rest of the group; however, the identification of the creator cannot be performed without authentication from the rest of the members of the group.

Tools used in network forensics.

Packet Capture & Analysis:

Wireshark: The industry standard for in-depth, interactive protocol analysis.

Tcpdump/Tshark: Command-line packet sniffers for live capture and creating PCAP files.

Arkime (formerly Moloch): Large-scale, indexed packet capture and search.

Network Security Monitoring (NSM) & Intrusion Detection (IDS):

Zeek (formerly Bro): Converts traffic into high-level, actionable logs.

Snort: An open-source IDS/IPS used to detect malicious traffic.

Suricata: A high-performance IDS/IPS/NSM engine.

Forensic Analysis & Extraction:

NetworkMiner: A tool designed for automated artifact extraction (files, images, credentials) from packet captures.

NetWitness Investigator: Used for deep packet inspection and session reconstruction.

Log Analysis & SIEM:

Splunk/ELK Stack: Used for analyzing logs from firewalls, routers, and IDS.

Integrated Forensic Platforms:

Security Onion: A Linux distro for threat hunting that integrates Snort, Zeek, and Wazuh.

Kali Linux: A comprehensive platform containing numerous pre-installed forensic tools.